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Exam Questions

Section 1, Module 1: Reading and Writing

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Question 1

The following text is adapted from Jhumpa Lahiri's 2003 novel *The Namesake*. Ashima and her husband, Ashoke, are traveling from the United States to India.

Ashoke puts his hand over hers as the plane begins to gather speed. And then Boston tilts away and they ascend effortlessly over a blackened Atlantic.

©2003 by Jhumpa Lahiri

As used in the text, what does the word "gather" most nearly mean?

A) Gain

B) Doubt

C) Welcome

D) Force

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Question 2

The _____blank Wendy Wan-Long Shang's book *The Great Wall of Lucy Wu* has been very positive. Many reviewers have praised the book, and it won the Asian/Pacific American Award for Literature.

Which choice completes the text with the most logical and precise word or phrase?

A) response to

B) neglect of

C) confusion about

D) inspiration for

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Question 3

Microbiologist Radamés Cordero and team investigated surface temperatures of a variety of fungi, including *Amanita muscaria* and a species from the genus *Cantharellus*, in a study _____blank a previous observation: Cordero had taken thermal images of various wild mushrooms while hiking and found that the mushrooms all appeared to be colder than the surrounding environment, raising questions about thermoregulation in fungi.

Which choice completes the text with the most logical and precise word or phrase?

A) released in

B) acknowledged in

C) excluded by

D) prompted by

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Question 4

Research on *Chaenocephalus aceratus* and other icefishes of the Teleostei class has shown that their hearts are pale due to low levels of myoglobin expression, which was once thought to be related to their polar habitats. Recently, researchers have found that some tropical teleost species (such as *Sphaeramia nematoptera*) also have pale hearts, raising the possibility that myoglobin deficiency in teleosts could be _____blank habitat.

Which choice completes the text with the most logical and precise word or phrase?

A) irrespective of

B) mediated by

C) predictive of

D) obscured by

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Question 5

In 2013 Bridie J.M. Allan and colleagues published a study concluding that ocean acidification has a strong effect on the behavior of *Pomacentrus amboinensis*, a species of fish. However, Allan and colleagues' study relied on a mean sample size of only about 18 fish. In a 2022 review of various scientists' conclusions about the impacts of ocean acidification on fish behavior, Josefin Sundin and colleagues caution that relying on such a relatively small sample size can increase the potential for biased analysis. Such analysis, in turn, can contribute to reports of exaggerated effects.

Which choice best describes the function of the underlined sentence in the text as a whole?

A) It introduces a detail about Allan and colleagues' study that is addressed in the sentences that follow.

B) It offers an alternative explanation for the behavior of *Pomacentrus amboinensis* that is described in the previous sentence.

C) It emphasizes the threat of ocean acidification to *Pomacentrus amboinensis* that is mentioned in the previous sentence.

D) It states the conclusion reached by Allan and colleagues that is elaborated on in the sentences that follow.

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Question 6

Editor Jared Shurin's 2023 anthology *The Big Book of Cyberpunk* contains 108 stories, an enormous number when compared to the 1986 cyberpunk anthology *Mirrorshades*, which has only twelve. Encompassing both early (e.g., Fritz Leiber's 1950 story "Coming Attraction") and more recent (e.g., Erica Satifka's 2021 story "Act of Providence") contributions, the broad chronological scope of Shurin's volume may make it seem more comprehensive than *Mirrorshades*, but *Mirrorshades*'s focus is more carefully considered, rendering it the more authoritative anthology of this science fiction subgenre.

Which choice best describes the function of the underlined portion in the text as a whole?

A) It supports the assertion that Leiber and Satifka should be considered essential figures in the cyberpunk movement.

B) It indicates a sensible perception of a cyberpunk anthology that can nonetheless be disputed for a good reason.

C) It suggests that, because they are relatively up to date, more recently published collections of cyberpunk stories are likely to be more inclusive than earlier anthologies.

D) It helps explain why two compilers of anthologies have different ideas regarding which short stories should be labeled as "cyberpunk."

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Question 7

Wenheng Zhang and colleagues have explored how convergent evolution—a phenomenon that occurs when the same trait evolves independently in two reproductively separate lineages—can result from a genetic mechanism shared by both lineages. Meanwhile, Patricia J. Wittkopp and colleagues have investigated how convergence occurs through different genetic mechanisms, but the relative prevalence of convergence through shared and different genetic processes is still poorly understood. This motivated biologists Delbert A. Green II and Cassandra G. Extavour to evaluate both types of convergence in a single study for their 2012 paper.

Which choice best states the function of the underlined portion in the text as a whole?

A) It clarifies a concept that the author implies was unclear in the studies mentioned in the text.

B) It provides examples of how a phenomenon was studied by scientists in the field before Green and Extavour's study.

C) It introduces a method of scientific analysis that is discussed in greater detail later in the text.

D) It gives a basic description of a phenomenon that is central to the discussion that follows.

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Question 8

Text 1

As part of a study of ungulates (hoofed mammals) native to the Yellowstone River basin, fresh dung samples from identified individuals were collected in the Confederated Salish and Kootenai Tribes Bison Range of Montana. Based on steroidal analysis, on average the bison and elk samples had similar levels of beta-sitosterol and stigmastanol, which distinguished these species' profiles from that of moose.

Text 2

Although fecal-steroidal profile alone is not sufficient to distinguish between samples from bison and those from elk, the relative levels of zoostanols (steroids with animal origins), such as coprostanol and 24-ethyl-coprostanol, in sediments at Buffalo Ford Lake in Yellowstone National Park show that bison, elk, or both were the predominant ungulates in the lake's watershed for all strata of sediment examined, going as far back as 2,300 years.

Which conclusion about steroid profiles is supported by information in both Text 1 and Text 2?

A) The profiles for bison and elk contain similar concentrations of various steroids.

B) Each of the ungulate species in the Buffalo Ford Lake watershed has a distinctive and unique steroidal profile.

C) Some steroids in the bison and elk profiles are more likely than others to be preserved for long time spans.

D) The profiles of ungulates in and around the Yellowstone basin show that bison and elk have historically been more prevalent than moose in the area.

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Question 9

Did you know that the *Warwick* is just one of around three million known shipwrecks spread across the world's oceans? Leila Hamdan and team wanted to find out how wooden shipwrecks affect the distribution of microbes on the seafloor. The researchers studied two wooden shipwrecks in the Gulf of Mexico. They placed pieces of pine and oak between 0 and 200 meters away from each shipwreck to collect samples of microbes. They found that pine and oak pieces placed about 125 meters from the shipwrecks had the most species of microbes on them.

According to the text, at what approximate distance from the studied shipwrecks did the researchers observe the most species of microbes on the pine and oak pieces?

A) 200 meters

B) 0 meters

C) 30 meters

D) 125 meters

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Question 10

According to the text, where was the National Association of Black Journalists founded?

A) Philadelphia

B) Detroit

C) Washington, DC

D) San Diego

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Question 11

Baleen whales eat up to 30 percent of their total body mass in krill (tiny shrimplike creatures) per day. So in one day, a humpback whale weighing 30,000 kg could eat 9,000 kg, while a southern right whale weighing 80,000 kg could ingest a whopping 24,000 kg. Over the last century, baleen whale populations have declined, and contrary to some scientists' expectations, so have krill populations. Matthew S. Savoca and colleagues resolve this apparent discrepancy by pointing out that baleen whales cycle iron in the ocean, helping support phytoplankton populations, which, in turn, sustain krill populations.

Based on the text, what is one likely effect of the change in baleen whale populations over the last century?

A) The maximum body sizes of humpback and southern right whales have increased over the last century.

B) The quantity of krill needed to support a single southern right whale has increased over the last century.

C) The iron richness of the krill that baleen whales eat has declined over the last century.

D) Populations of phytoplankton that krill feed on have declined over the last century.

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Question 12

Average Hours Worked per Person per Year in 1950 and 2017

Country	1950	2017	Change in hours	Percent change in hours
Brazil	2,042	1,709	−333	−16%
Japan	2,030	1,738	−292	−14%
Switzerland	2,040	1,590	−450	−22%
Germany	2,427	1,354	−1,074	−44%

Calculations may be inexact due to rounding.

A student in an economics course is examining the decline since 1950 in average hours worked per person per year in various nations due to both increased productivity and the adoption of policies that limit working hours. The first task in this investigation is to determine how the decline in Switzerland compares to that in other countries. The student finds that _____blank

Which choice most effectively uses data from the table to complete the statement?

A) though the percent decrease in hours worked in Switzerland was less than that in Germany, it was greater than that in Brazil and Japan.

B) the decline in number of hours worked was greater in Switzerland than it was in Germany, Brazil, or Japan.

C) while the number of hours worked rose in Switzerland from 1950 to 2017, it declined in Germany, Brazil, and Japan.

D) though the percent decrease in hours worked in Switzerland was greater than that in Germany and Japan, it was less than that in Brazil.

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Question 13

The Age of Innocence is a 1920 novel by Edith Wharton set in New York City in the 1870s. In the novel, Newland Archer attends an opera. Newland compares his intellect favorably to that of other men of New York City society who are in the audience: _____blank

Which quotation from *The Age of Innocence* best illustrates the claim?

A) "It surprised [Newland] that life should be going on in the old way when his own reactions to it had so completely changed."

B) "Newland Archer felt himself distinctly the superior of these chosen specimens of old New York gentility; he had probably read more, thought more, and even seen a good deal more of the world, than any other man of the number."

C) "No expense had been spared on the setting, which was acknowledged to be very beautiful even by people who shared [Newland's] acquaintance with the Opera houses of Paris and Vienna."

D) "Singly [the men around Newland] betrayed their inferiority; but grouped together they represented 'New York,' and the habit of masculine solidarity made him accept their doctrine on all the issues called moral."

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Question 14

In 1927, the Swedish Academy presented Henri Bergson of France with an _____blank that few authors would turn down: the Nobel Prize in Literature.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) award:

B) award

C) award,

D) award;

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Question 15

Paleontologist Amane Tajika and colleagues analyzed the shells of two marine mollusks called nautilids that were harvested from New Caledonia. The researchers found that samples from shell sections that formed before hatching, including sample F04, formed in 20–22°C water, whereas samples from sections that formed after hatching, including M27, formed in 13–15°C water. These findings are consistent with the temperatures Fijian nautilids encounter during their corresponding stages of development. Because water temperature varies with depth, a biology student hypothesized that the New Caledonian nautilids lay their eggs at approximately the same depth as Fijian nautilids do.

Which finding, if true, would most directly weaken the student's hypothesis?

A) Fijian nautilids generally live at lower depths than New Caledonian nautilids do, though Fijian nautilids migrate to the typical depth of New Caledonian nautilids to lay their eggs.

B) Once a nautilid hatches, it is a very mobile animal, so the temperature values reported by Tajika and colleagues for the adult nautilids are averages over a potentially large range of depths.

C) The water temperature at a given depth tends to be lower in waters near New Caledonia than it is at that depth in waters near Fiji.

D) There are few locations near Fiji with water temperatures of 13–15°C that are suitable for nautilid eggs.

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Question 16

The *Mammillaria* cactus *M. guerreronis* occurs naturally only in the state of Guerrero in Mexico, and the smallness of its range makes it especially vulnerable to extinction. The traditional single-species approach to conservation emphasizes the need to focus on individual species most at risk, like *M. guerreronis*, but recently, conservationists have argued that an ecosystem-based approach that incorporates the many interactions between the climate, terrain, and various species of a given geographical area may lead to better outcomes for all the species in a given location. An ecosystem-based approach to the conservation of *M. guerreronis* could thus _____blank

Which choice most logically completes the text?

A) allow conservationists to better consider how climatic changes affecting Guerrero may reduce the number of species competing with *M. guerreronis*.

B) identify conservation strategies that may not be exclusively aimed at protecting *M. guerreronis* but that benefit the species nonetheless.

C) lead to a better understanding of how the distribution of *Mammillaria* species throughout Mexico has affected their survival.

D) be unable to evaluate whether *M. guerreronis* would be capable of surviving in habitats outside of Guerrero.

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Question 17

When first looking at San Carlos Apache artist Evalena Henry's woven baskets, viewers are likely to notice the baskets' detailed animal imagery. Upon closer inspection, Henry's careful construction _____blank apparent.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) have become

B) are becoming

C) becomes

D) become

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Question 18

The Hereford map is a mappa mundi from the fourteenth century. Currently held at Hereford Cathedral in Hereford, England, it is one of over 1,000 such world maps from the European Middle Ages still remaining, with the even older Vercelli map from the thirteenth century (now held at the Archivio Capitolare in Vercelli, Italy) _____blank another one.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) was

B) being

C) is

D) has been

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Question 19

When documenting butterfly diversity at Rupa Lake in Nepal, researcher Alyssa B. Stewart and her colleagues spotted a very rare butterfly of the Hesperiidae family and, after careful examination, knew exactly what species it belonged _____blank *Ochus subvittatus*.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) to and

B) to:

C) to;

D) to

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Question 20

The relationship between grassland arthropod populations and agricultural fertilizers containing macronutrients—such as nitrogen, phosphorus, and potassium—was the focus of a 2013 study in the _____blank by the researcher Laura A. Burkle, the study examined a variety of arthropod specimens, which had been collected by hand.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) United States, led

B) United States and led

C) United States led

D) United States. Led

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Question 21

The World Cup of men's soccer, one of the biggest sporting events on the planet, brought 32 national teams from six continents to the host country, Qatar, in 2022. _____blank only 16 teams, most from Europe and the Americas, competed in the 1954 World Cup in Switzerland.

Which choice completes the text with the most logical transition?

A) In other words,

B) In contrast,

C) For instance,

D) Therefore,

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Question 22

Alabama is one of several US states to enact statutes supporting the incorporation of worker cooperatives, businesses with democratic governance. Many scholars testify to the economic sustainability of the co-op model while acknowledging its challenges. _____blank those who declare the model inherently unviable can discount documented successes, while those who champion it may understate its challenges.

Which choice completes the text with the most logical transition?

A) That said, some experts overlook such nuance;

B) Nevertheless, how these challenges might be overcome remains contentious;

C) Ultimately, these complex issues require complex solutions;

D) Thus, the scholars are skeptical about the co-op model's viability;

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Question 23

Until 1992, most Olympic Games included demonstration sports, which did not award medals. _____blank a demonstration of glíma, a traditional Nordic form of wrestling, was featured at the 1912 Summer Olympics in Stockholm.

Which choice completes the text with the most logical transition?

A) For example,

B) Meanwhile,

C) Someday,

D) On the other hand,

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Question 24

Many places that gained their independence from Spain in the nineteenth century still bear the legacy of the Spanish Empire in their language, including Spanish-speaking Cuba, which left the empire in 1898. _____blank which ceased to be part of the Spanish Empire in 1648; Spanish is rarely spoken in the Netherlands in the twenty-first century.

Which choice completes the text with the most logical transition?

A) This contrast is evident in the Netherlands,

B) Another example is the Netherlands,

C) The reverse is true of the Netherlands,

D) The results can be heard in the Netherlands,

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Question 25

While researching a topic, a student has taken the following notes:

- Elizabeth Catlett (1915–2012) was a celebrated African American artist.
- She is best known for creating sculptures and prints that explore the Black experience.
- Catlett's sculpture *Sister* is made of bronze.
- She created this sculpture in 1973.

The student wants to specify the year that Elizabeth Catlett created the sculpture *Sister*. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) Elizabeth Catlett, a celebrated artist, was born in 1915.

B) Celebrated artist Elizabeth Catlett is best known for creating sculptures and prints that explore the Black experience.

C) Elizabeth Catlett was a celebrated artist who created prints as well as sculptures.

D) Elizabeth Catlett created the bronze sculpture *Sister* in 1973.

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Question 26

While researching a topic, a student has taken the following notes:

- NASA mission Expedition 14 took place from September 18, 2006, to April 21, 2007.
- Astronauts Michael López-Alegría and Sunita Williams were crew members of Expedition 14.
- January 31, 2007: López-Alegría and Williams performed an extravehicular activity (EVA) to replace instruments on the Hubble Space Telescope.
- An EVA (also known as a spacewalk) is an activity done by astronauts outside the spacecraft.
- The duration of their EVA was 7 hours and 55 minutes.
- Theirs is counted among the longest completed spacewalks to date.

The student wants to emphasize the duration of López-Alegría and Williams's EVA. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) López-Alegría and Williams completed an EVA, also known as a spacewalk, in which they left the spacecraft to replace instruments on the Hubble Space Telescope.

B) Lasting 7 hours and 55 minutes, López-Alegría and Williams's EVA is one of the longest spacewalks completed to date.

C) During their time as crew members of Expedition 14, López-Alegría and Williams completed an EVA.

D) The EVA completed by Expedition 14 crew members López-Alegría and Williams took place from September 18, 2006, to April 21, 2007.

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Question 27

While researching a topic, a student has taken the following notes:

- Cheryll is the nickname of a *Triceratops* fossil specimen housed at the Palm Beach Museum of Natural History.
- The Palm Beach Museum of Natural History is located in Wellington, Florida.
- Cheryll lived in the Late Cretaceous period.
- Dio is the nickname of a *Triceratops* fossil specimen housed at the Royal Ontario Museum.
- The Royal Ontario Museum is located in Ontario, Canada.
- Dio lived in the Late Cretaceous period.

The student wants to contrast the locations of the two specimens. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) Cheryll is housed at the Palm Beach Museum of Natural History in Wellington, Florida, while Dio is housed at the Royal Ontario Museum in Ontario, Canada.

B) The *Triceratops* fossil specimen Cheryll is not the only such specimen currently housed in a museum or institute.

C) The Royal Ontario Museum in Ontario, Canada, houses Dio, a *Triceratops* fossil specimen.

D) Like Cheryll, Dio is a *Triceratops* fossil specimen that lived in the Late Cretaceous period.

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Section 1, Module 2: Reading and Writing

Question 1

In addition to the giant stock exchanges in places like London, Mumbai, and Tokyo, there are many smaller exchanges such as the Beirut Stock Exchange in Beirut, Lebanon. Though small, by being dedicated to Lebanon, it can better address factors that help local businesses _____blank, and that focus on the success of local businesses benefits investors and the local economy.

Which choice completes the text with the most logical and precise word or phrase?

A) thrive

B) disperse

C) ebb

D) overspend

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Question 2

The following text is from Rudolfo Anaya's 1972 novel *Bless Me, Ultima*. Two of the narrator's adult brothers have decided to move away from home. Another brother has decided not to accompany them.

In the morning León and Eugene were gone, but Andrew remained. They had talked long into the night, and finally he had given up the idea of going with them. I think he did not like to follow their ways, and he wanted to please my mother.

©1972 by Rudolfo Anaya

As used in the text, what does the word "follow" most nearly mean?

A) Justify

B) Imitate

C) Mention

D) Restore

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Question 3

Which choice completes the text with the most logical and precise word or phrase?

A) waste

B) advice

C) positions

D) rewards

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Question 4

Corrine Walsh and colleagues tested whether soil microorganisms from ecologically different settings—aspen grove, irrigated pasture, and so on—affect plants' flavor chemistry. Although they found associations between plants' glucosinolates (compounds that produce spicy and bitter flavors) and plants' growth conditions, variations in glucosinolates could not be _____blank soil microbes specifically.

Which choice completes the text with the most logical and precise word or phrase?

A) forestalled by

B) subverted by

C) attributed to

D) excluded from

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Question 5

The Walt Whitman House in Camden, New Jersey, is the house where Walt Whitman lived for eight years and where he wrote later editions of his poetry collection *Leaves of Grass*. Now the house is a museum dedicated to Whitman's life and writings. Visitors travel to this museum to feel more connected to an author they admire. To encourage that sense of connection, the museum's curators have set up exhibits with furnishings and décor owned by Whitman. Such exhibits help visitors understand day-to-day life in the house where Whitman felt inspired.

Which choice best states the main purpose of the text?

A) To describe a museum focused on Whitman's life and works

B) To explain why readers enjoy Whitman's works

C) To summarize Whitman's biography

D) To compare the Walt Whitman House with other museums

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Question 6

The following text is adapted from Henrik Ibsen's 1899 play *When We Dead Awaken* (translated by William Archer in 1907). Ulfheim is introducing himself to Professor Rubek, a sculptor.

ULFHEIM: I believe I have the honour of addressing no less a person than the great Sculptor Rubek.

PROFESSOR RUBEK: (*Nods.*) I remember meeting you once or twice—the autumn when I was last at home.

ULFHEIM: That's many years ago, now. And then you weren't so illustrious as I hear you've since become.

Which choice best describes the function of the underlined portion in the text?

A) It reveals that Professor Rubek first met Ulfheim many years ago.

B) It emphasizes that Professor Rubek has a distinguished reputation.

C) It suggests that Ulfheim prefers autumn to spring, summer, or winter.

D) It indicates that many other sculptors are more famous than Professor Rubek.

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Question 7

Text 1

Brown University and the investment bank Barclays are two of the many institutions offering training programs in entrepreneurship. But what results do such programs produce? In a study of business school graduates in Norway, researchers Lars Kolvereid and Øystein Moen addressed this question and convincingly demonstrated that participants showed strong intentions to become entrepreneurs as a result of receiving entrepreneurial training.

Text 2

While studies of entrepreneurial training typically report positive results, a close look at these studies reveals widespread methodological shortcomings. For instance, a 1997 study by Lars Kolvereid and Øystein Moen found benefits of entrepreneurial education for graduates of Norwegian business schools, but the study only collected data on subjects after the training, not before, making it unclear whether the training actually had an effect.

Based on the texts, the author of Text 1 and the author of Text 2 would most likely disagree about the answer to which question?

A) Is the cost of conducting entrepreneurial training programs typically offset by the programs' benefits?

B) Did Kolvereid and Moen's study provide persuasive evidence that its results are attributable to the entrepreneurial training participants received?

C) Did Kolvereid and Moen report that participants who received the entrepreneurial training showed strong intentions to become entrepreneurs?

D) Are there convincing experimental results showing that different types of entrepreneurial training programs are associated with different outcomes?

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Question 8

Scientists study the abundance and the activity ratios (AR)—the relative radioactive decay rates of two isotopes—of uranium (U) isotopes in bodies of water to track sediment transport and deposition and to monitor groundwater quality. However, isotope abundance and AR are not always uniform across a single body of water, which may affect the reliability of results if sampling is not sufficiently representative. Chen Zhao et al. analyzed concentrations of the isotope U-238 and the U-234/U-238 AR at multiple depths and locations in two lakes in China, finding no significant variation in values within the same lake.

Information in the text best supports which statement about Zhao et al.'s findings?

A) They illustrate why scientists investigating sediment transport and deposition and groundwater quality ought to evaluate how evenly U abundance and AR are distributed in the bodies of water they are examining.

B) They challenge the assumption that U abundance and AR sometimes vary from lake to lake, since the U-238 concentration and U-234/U-238 AR were consistent across the sampled lakes.

C) They indicate that researchers performing isotopic analysis in the lakes Zhao et al. studied can likely rely on localized sampling within each lake, since values from any one location or depth are representative of the entire lake.

D) They demonstrate that the number of samples Zhao et al. collected from the two lakes is sufficient to yield equally reliable information about sediment transport and deposition and groundwater quality in those two lakes.

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Question 9

One challenge faced by researchers studying global urbanization is that different countries have different definitions of what constitutes an urban settlement. Many countries define urban settlements based on the number of people living in them. However, countries do not all use the same number; for example, Greenland uses a minimum population of 200, while Benin uses 10,000. Other countries use a combination of population and other factors, like how close buildings are to each other. This variation makes it difficult for researchers to compare aspects of urbanization in different countries.

Which choice best states the main idea of the text?

A) It is difficult for researchers to compare aspects of urbanization globally because of the various ways different countries define urban settlements.

B) Even though countries use different methods to define urban settlements, researchers consider any area having a population of at least 10,000 to be urban.

C) Researchers focus on factors other than population when conducting studies on urbanization in Greenland.

D) Urbanization is easily compared between countries when they have similar total populations.

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Question 10

Brown Bears in Katmai National Park, Alaska

Bear identification number	Sex	Age (years)	Approximate weight (pounds)
122	male	3	200
117	female	6	325
157	male	7	450
123	female	11	350

Scientists collected information about brown bears in Katmai National Park in Alaska. This information included each bear's sex, age, and approximate weight. Among the bears whose information is shown in the table, the youngest one was the bear with identification number _____blank

Which choice most effectively uses data from the table to complete the statement?

A) 123.

B) 157.

C) 117.

D) 122.

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Question 11

Year That Foreign Investors Were First Allowed to Purchase Shares

Country	Year
India	1986
Morocco	1988
Indonesia	1989

In the 1980s and 1990s, many countries began allowing foreign investors to purchase shares of companies based in those countries.

According to the table, in what year did India make the change mentioned in the text?

A) 1981

B) 1998

C) 1983

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D) 1986

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Question 12

Psychologists Gregory Bryant, Dorsa Amir, and colleagues investigated cross-cultural perceptions of spontaneous (real) laughter and volitional (fake or forced) laughter. Study participants from 21 societies, including those in Japan and Turkey, listened to randomized recordings of 18 spontaneous laughs taken from natural conversations between pairs of women and 18 volitional laughs produced separately by 18 different women in response to an instruction to laugh. Analysis of the participants' evaluations of the laughs prompted the team to conclude that the ability to distinguish between spontaneous and volitional laughter appears to be universal across cultures.

Which potential finding from another study, if true, would most directly strengthen the team's conclusion?

A) Listeners from both Japan and Turkey tend to be able to determine how well two people know each other based on recordings of their conversation.

B) Spontaneous laughter typically has a recognizably higher pitch than volitional laughter, regardless of what society the speakers in a conversation are from.

C) There is some evidence of correlation between the average size of communities in a society and judgments of spontaneous and volitional laughter by listeners in that society.

D) Acoustic features of spontaneous laughter, such as pitch and intensity, have measurable variations both within and across societies.

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Question 13

Certain species, such as Hay's Spring amphipods (*Stygobromus hayi*), lack functional eyes or ocular organs altogether and rely on other mechanisms for navigation. Licheng Yuan and colleagues investigated such mechanisms in the Chinese red-headed centipede (*Scolopendra subspinipes mutilans*), which has no organs with photoreceptors (light-detecting cells) but can sense and avoid sunlight. The researchers placed individual specimens in a clear container (container 1) connected to a container covered in black tape (container 2) and then exposed the containers to simulated sunlight. They observed a rapid increase in the temperature of the centipedes' antennae with the onset of exposure, followed by the centipedes' escape to container 2. Based on their observations, the researchers concluded that thermal receptors in the antennae are likely responsible for light detection in the centipedes.

Which additional finding from Yuan and colleagues' study, if true, would best support the researchers' conclusion?

A) When body segments other than the centipedes' antennae were monitored, the temperature of these body segments were found to increase in the presence of sunlight.

B) The centipedes showed the same tendency to move to container 2 when simulated sunlight was accompanied by artificially reduced temperatures.

C) Chinese red-headed centipedes are more proficient at detecting light than other centipede species are.

D) When their antennae were covered with a reflective material, the centipedes remained exposed in container 1 for longer durations.

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Question 14

Introducing a new species into a delicate ecosystem can often be harmful. The common waxbill and other fruit-eating birds have established themselves on the Hawaiian island of Oahu after being brought there in the last 150 years. Many plant species on the island rely on the activity of fruit-eating birds to maintain and increase their populations. In recent studies, common waxbills have been found to spread seeds of native forest plants that are at risk of extinction, such as the olomea shrub and the alahee tree, suggesting that _____blank

Which choice most logically completes the text?

A) non-native birds, such as the common waxbill, are unlikely to eat fruits from non-native forest plants on Oahu.

B) non-native birds have had both positive and negative effects on Oahu's native plants.

C) although non-native birds initially damaged the plants in Oahu's ecosystem, they are now proving useful to them.

D) rather than harming Oahu's ecosystem, some non-native birds provide an important benefit on Oahu.

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Question 15

Many Indigenous songs serve as repositories of ecological knowledge—e.g., Yi songs about the natural environment and O'odham songs about desert plants. In a study by ethnobiologist Dana Lepofsky et al., a song keeper for the Kwakwaka'wakw people provided access to ancient songs referring to the people's creation of terraced intertidal gardens to foster clams for consumption. Cross-references of that information with both archaeological and paleoecological records have revealed significant increases in clam size and productivity concurrent with the most prevalent use of these clam gardens. This finding suggests that researchers should consider _____blank

Which choice most logically completes the text?

A) the strong implication that the beneficial effects are primarily attributable to ecological factors unrelated to the activities of the Kwakwaka'wakw people.

B) the possibility that other songs of other Indigenous peoples transmit ecological information that has useful practical applications.

C) the reliability of archaeological and paleoecological evidence that often underpins conclusions drawn from Indigenous songs.

D) whether the Kwakwaka'wakw people originated the method of cultivation detailed in the songs or adapted methods observed among other groups of Indigenous peoples.

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Question 16

The smoke point of corn oil _____blank around 450 degrees Fahrenheit. When corn oil is heated past this temperature, it will begin to combust and emit smoke.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) is—

B) is;

C) is

D) is,

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Question 17

Because many of the first cattle ranchers in the US learned the trade from Spanish-speaking Mexican *vaqueros* (cowboys), many ranching terms in the English language come from Spanish. One example is the word "lasso." It comes from the Spanish word *lazo*. Another example is "rodeo." _____blank comes from the Spanish word *rodear*.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) It

B) Someone

C) She

D) He

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Question 18

Located in what is now Egypt, the Pyramid of Userkaf is one of the oldest known structures in the world, _____blank researchers a glimpse into ages long gone.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) to provide

B) provides

C) provide

D) providing

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Question 19

In the Suzuki method, folk songs like “Mary Had a Little Lamb” are taught to beginning music students as young as three, while students of traditional methods often only _____blank such songs at the age of five.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) begin learning

B) beginning to learn

C) having begun to learn

D) to begin learning

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Question 20

Marché Monge has operated for more than a century, making _____blank both a popular shopping destination and an important piece of Paris's history.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) those

B) it

C) these

D) one

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Question 21

When a soil is classified using the International Union of Soil Sciences' system, its similarity to each of the 32 soil reference groups—defined by several criteria—_____blank evaluated, one group at a time, until a match is found. The comparisons are made in a specific order; thus, the criteria for kastanozems (soils with highly organic topsoil) are only applied if the soil doesn't match the criteria for planosols.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) are

B) was

C) have been

D) is

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Question 22

In the early 1920s, researchers at the US Geological Survey determined the geographic center of Maine using a simple method. First, they attached a string to the middle point of a cardboard cutout of the state and suspended it in the air. _____blank they adjusted the suspension point until the cutout hung perfectly flat. Ultimately, this point was determined to be the center.

Which choice completes the text with the most logical transition?

A) Then,

B) In other words,

C) Instead,

D) First of all,

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Question 23

When the first railroads were built in Australia in the 1800s, a variety of track sizes (gauges) were used across the continent, resulting in what historians call the “railway gauge muddle.” The Mornington Railway line in Victoria used broad tracks. _____blank the Rockhampton Steam Tramway line in Queensland used narrow gauge tracks.

Which choice completes the text with the most logical transition?

A) In sum,

B) By contrast,

C) Similarly,

D) For instance,

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Question 24

While researching a topic, a student has taken the following notes:

- Snow hydrologist Zoe Wellschlager helps maintain automated snow measurement stations throughout the western United States.
- They are called SNOTEL sites.
- There are around 90 sites throughout Wyoming.
- There is one at Cottonwood Creek.

The student wants to emphasize the number of SNOTEL sites in Wyoming. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) In Wyoming alone, there are around 90 SNOTEL sites located throughout the state.

B) One particular SNOTEL site is located at Cottonwood Creek in Wyoming.

C) SNOTEL sites collect snow measurement data throughout the state of Wyoming.

D) Snow hydrologist Zoe Wellschlager helps maintain SNOTEL sites, such as the one at Cottonwood Creek, Wyoming.

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Question 25

While researching a topic, a student has taken the following notes:

- Wind is a renewable energy source.
- A wind farm uses turbines to convert wind into electrical power.
- A wind farm's maximum electrical output is called its nameplate capacity.
- Hornsdale Wind Farm is located in Hornsdale, Australia.
- Its nameplate capacity is 316 megawatts.

The student wants to emphasize the location of Hornsdale Wind Farm. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) Hornsdale Wind Farm has a nameplate capacity of 316 megawatts.

B) Nameplate capacity is another name for a wind farm's maximum electrical output.

C) Hornsdale, Australia, is home to Hornsdale Wind Farm.

D) Turbines can be used to generate electricity from a renewable energy source: the wind.

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Question 26

While researching a topic, a student has taken the following notes:

- The Wayang Windu geothermal power plant in Indonesia uses a flash steam system to generate electricity.
- Flash steam technology requires geothermal reservoir temperatures above 180°C.
- The use of flash steam systems worldwide is limited because many geothermal reservoirs aren't hot enough.
- The Raft River geothermal power plant in Idaho uses a binary cycle system.
- Binary cycle technology can generate energy from lower-temperature (less than 180°C) geothermal reservoirs.
- Binary cycle systems involve higher maintenance costs than flash steam systems.

The student wants to compare the disadvantages of the geothermal systems used at the Wayang Windu and Raft River plants. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) Compared with the system used at the Wayang Windu plant, the system used at the Raft River plant has a notable disadvantage: its cost.

B) Unlike the Raft River plant, the Wayang Windu plant uses a flash steam system, which requires a less common resource: geothermal reservoirs with temperatures above 180°C.

C) The system used at the Raft River plant overcomes the temperature limitations of the Wayang Windu plant's system but is more costly to maintain.

D) The system at the Wayang Windu plant requires temperatures above 180°C, while the system at the Raft River plant can operate at lower temperatures.

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Question 27

While researching a topic, a student has taken the following notes:

- *ArcelorMittal Orbit* is an artwork by Sir Anish Kapoor and Cecil Balmond.
- It is a 375-foot steel tower with a spiraling staircase and working slide.
- It is on display in a public park in London, England.
- *Čumil the Sewer Worker* is a work of art by Viktor Hulík.
- It is a bronze sculpture of a man emerging from under a manhole cover.
- It is on display on a public street in Bratislava, Slovakia.

The student wants to emphasize a similarity between *ArcelorMittal Orbit* and *Čumil the Sewer Worker*. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) *ArcelorMittal Orbit* and *Čumil the Sewer Worker* are both on display at public locations.

B) *ArcelorMittal Orbit*, but not *Čumil the Sewer Worker*, can be found in London, England.

C) Both *ArcelorMittal Orbit* and *Čumil the Sewer Worker* were created by Sir Anish Kapoor and Cecil Balmond.

D) *Čumil the Sewer Worker*, a bronze sculpture of a man emerging from under a manhole cover, is on display on a public street.

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Question 1

A quadratic function models the height, in feet, of an object above the ground in terms of the time, in seconds, after the object was launched. The model estimates that the object was launched from an initial height of 0.00 feet above the ground and was at a maximum height of 207.36 feet above the ground 3.6 seconds after it was launched. The model also estimates that the object was at a height of 184.32 feet above the ground x seconds after it was launched. What is a possible value of x?

Correct Answer:

2.4

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Question 2

The function h is defined by $h(x) = 5(9x - 10)$. What is the value of $h(10)$?

A) 400

B) 10

C) 50

D) 90

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Question 3

A pet service that takes care of both cats and dogs can care for at most 43 animals. Which inequality represents this situation, where c is the number of cats and d is the number of dogs?

A) $c - d \leq 43$

B) $c - d \geq 43$

C) $c + d \geq 43$

D) $c + d \leq 43$

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Question 4

$$9x + 17 = 9x + k$$

In the given equation, k is a constant. The equation has infinitely many solutions. What is the value of k?

Correct Answer:

17

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Question 5

The *y*-intercept of the graph of $y = 6(1.85)^x$ in the *xy*-plane is $(0, y)$. What is the value of y?

Correct Answer:

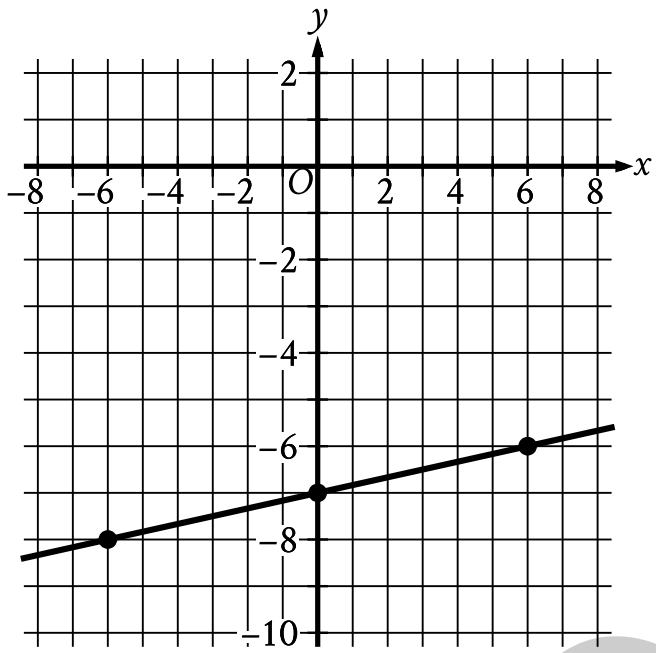
6

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Question 6



- The line slants gradually up from left to right.
- The line passes through the following points:
  - (negative 6 comma negative 8)
  - (0 comma negative 7)
  - (6 comma negative 6)

The graph of the linear function f is shown, where $y = f(x)$. What is the slope of the graph of f?

A) $\frac{1}{6}$

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B) -7

C) $-\frac{1}{6}$

D) 7

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Question 7

A triathlon is a multisport race consisting of three different legs. A triathlon participant completed the cycling leg with an average speed of 17.700 miles per hour. What was the average speed, in yards per hour, of the participant during the cycling leg?
$(1 \text{ mile} = 1{,}760 \text{ yards})$

Correct Answer:

31152

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Question 8

Triangle ABC is similar to triangle DEF, where angle A corresponds to angle D and angle C corresponds to angle F. Angles C and F are right angles. If $\tan A = \frac{612}{35}$, what is the value of $\sin D$?

A) $\frac{612}{613}$

B) $\frac{35}{613}$

C) $\frac{613}{612}$

D) $\frac{613}{35}$

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Question 9

The function $A(b) = 7b^2$ gives the area of a right triangle, in square centimeters (cm^2), if the length of its base is b cm and the length of its height is 14 times the length of its base. Which of the following is the best interpretation of $A(18) = 2{,}268$?

A) If the length of the base of the triangle is $2{,}268$ cm, then the length of the height of the triangle is 18 cm.

B) If the length of the base of the triangle is 18 cm, then the length of the height of the triangle is $2{,}268$ cm.

C) If the length of the base of the triangle is 18 cm, then the area of the triangle is $2{,}268$ cm^2.

D) If the length of the base of the triangle is $2{,}268$ cm, then the area of the triangle is 18 cm^2.

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Question 10

$$-4(v-9)(v-2)(v+7)=0$$

What value of v is a solution to the given equation?

A) -2

B) 7

C) 9

D) -4

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Question 11

Circle K has a radius of 8 millimeters (mm). Circle L has an area of 144π mm^2. What is the total area, in mm^2, of circles K and L?

A) 40π

B) 20π

C) 80π

D) 208π

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Question 12

$$x^2 + 49x + 1 = 0$$

What is one of the solutions to the given equation?

A) $\frac{-49+\sqrt{(49)^2-4(1)(1)}}{2(1)}$

B) $\frac{-49+\sqrt{(49)^2-4(49)(49)}}{2(49)}$

C) $\frac{-1+\sqrt{(1)^2-4(1)(49)}}{2(1)}$

D) $\frac{-1+\sqrt{(1)^2-4(49)(1)}}{2(49)}$

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Question 13

A model estimates that the initial number of photons in an X-ray beam is 800 when the beam reaches the surface of a certain material. The model also estimates that when the beam passes through this material, the number of photons in the beam decreases by 50% for each 7 millimeters of the material the beam passes through. Which equation represents this model, where I is the estimated number of photons in the beam when the beam reaches a point x millimeters from the surface?

A) $I = 800(7)^{\frac{x}{2}}$

B) $I = 800(0.5)^{x}$

C) $I = 800(2)^{\frac{x}{7}}$

D) $I = 800(0.5)^{\frac{x}{7}}$

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Question 14

In the *xy*-plane, line k passes through the points $(3, 0)$ and $(0, 5)$. Which of the following is an equation of line k?

A) $3x - 5y = 15$

B) $5x + 3y = 15$

C) $3x + 5y = 15$

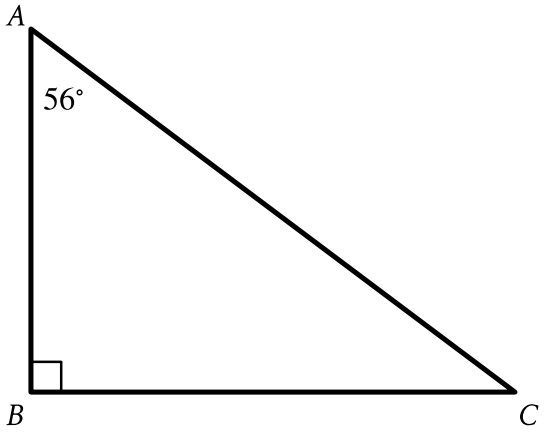
D) $5x - 3y = 15$

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Question 15



Note: Figure not drawn to scale.

- Angle upper B is a right angle.
- The measure of angle upper A is 56°.
- Note: Figure not drawn to scale.

In right triangle ABC shown, what is the measure of $\angle C$?

A) 124°

B) 146°

C) 56°

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D) 34°

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Question 16

$$x + 4y = -3$$
$$4x - y = 22$$

The solution to the given system of equations is (x, y). What is the value of $3x - 5y$?

Correct Answer:

25

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Question 17

A circle in the *xy*-plane has its center at $(-4, 1)$ and the point $(-8, 4)$ lies on the circle. Which equation represents this circle?

A) $(x-4)^2+(y+1)^2=5$

B) $(x+4)^2+(y-1)^2=25$

C) $(x-4)^2+(y+1)^2=25$

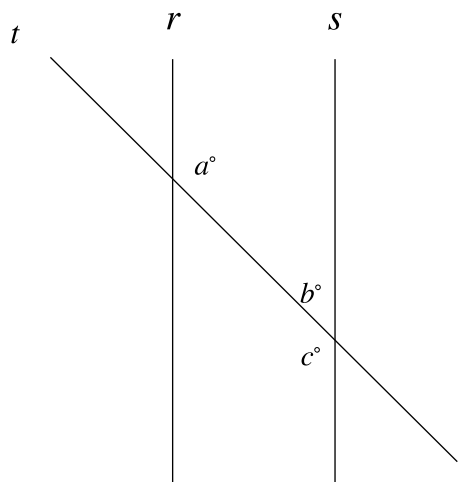
D) $(x+4)^2+(y-1)^2=5$

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Question 18



Note: Figure not drawn to scaleF

- Clockwise from top left, the 3 lines are labeled t, r, and s.
- Line t intersects both line r and line s.
- At the intersection of line t and line r, 1 angle is labeled clockwise from top left as follows:
  - Top right: a°
- At the intersection of line t and line s, 2 angles are labeled clockwise from top left as follows:
  - Top left: b°
  - Bottom left: c°
- A note indicates the figure is not drawn to scale.

In the figure, lines r and s are parallel and line t intersects both lines. If $a = 4k + 57$ and $b = \frac{k}{4} + 55$, what is the value of c?

Correct Answer:

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Question 19

$$0.5x + 1.2y = 9$$
$$1.5x + 3.6y = 27$$

How many solutions does the given system of equations have?

A) Infinitely many

B) Exactly two

C) Exactly one

D) Zero

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Question 20

For a study, a research team measured the heights of 380 king penguins from the Kerguelen Islands and 190 king penguins from the Falkland Islands. The research team determined that the mean height of the 380 king penguins from the Kerguelen Islands was 87 centimeters and the mean height of the 190 king penguins from the Falkland Islands was 93 centimeters. What was the mean height, in centimeters, of all 570 king penguins the research team measured for this study?

A) 91

B) 89

C) 90

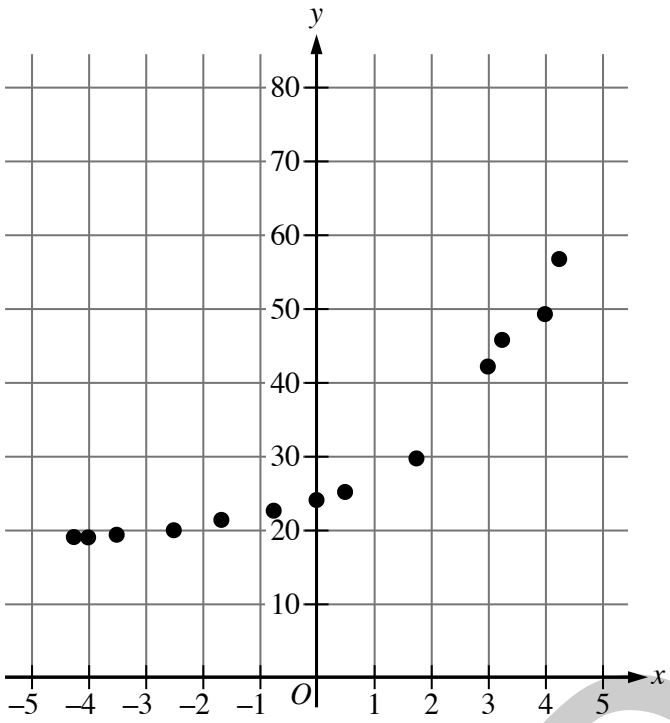
D) 88

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Question 21



- The scatterplot has 13 data points.
- The data points are in an exponential pattern trending up from left to right.
- The data points have the following approximate coordinates:
  - (negative 4.3 comma 19)
  - (negative 4 comma 19)
  - (negative 3.5 comma 19)
  - (negative 2.5 comma 20)
  - (negative 1.7 comma 21)
  - (negative 0.8 comma 23)
  - (0 comma 24)

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- (0.5 comma 25)
- (1.8 comma 30)
- (3 comma 42)
- (3.3 comma 46)
- (4 comma 49)
- (4.3 comma 57)

Which of the following equations is the most appropriate model for the data shown in the scatterplot?

A) $y = 24(1.55)^x$

B) $y = 24(1.55)^x + 18$

C) $y = (1.55)^x + 23$

D) $y = 6(1.55)^x + 18$

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Question 22

$$18qrt - 2qrs + 10rst = 0$$

In the given equation, q, r, s, and t are positive, and $\frac{q}{t}$ is greater than 5. Which expression is equivalent to s?

A) $\frac{-9qt}{q-5t}$

B) $\frac{9qt}{q-5t}$

C) $18qrt$

D) $\frac{2qt}{q-5t}$

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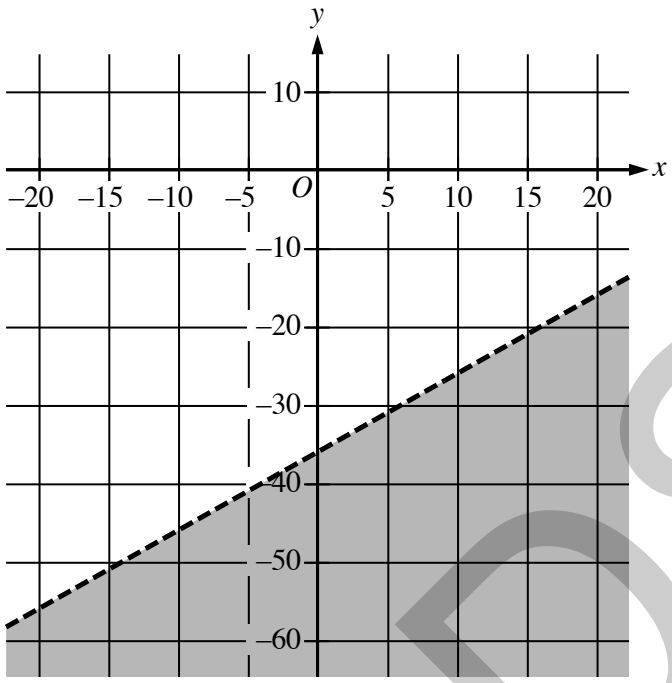
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Question 1

$$y > -36$$

Which shaded region represents the solutions to the given inequality?

A)



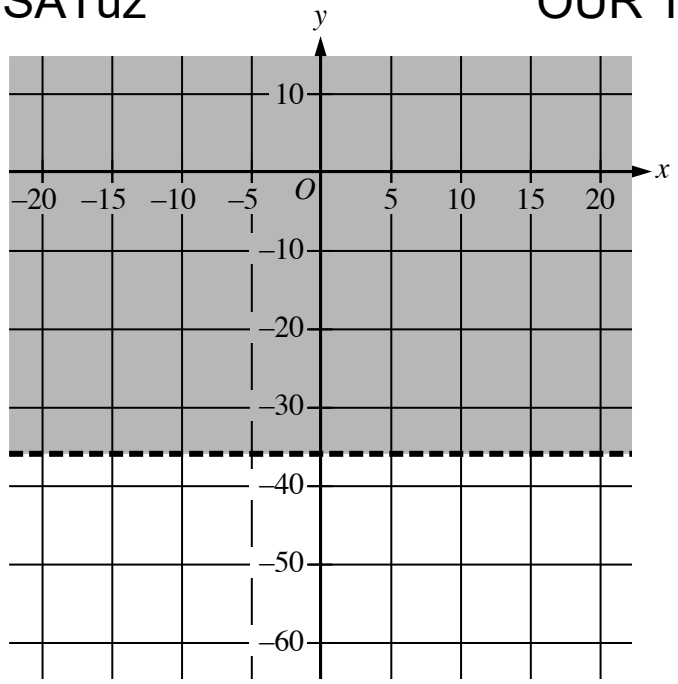
- The boundary of the inequality is a dashed line.
  - The line slants gradually up from left to right.
  - The line passes through the following points:
    - (negative 10 comma negative 46)
    - (0 comma negative 36)
    - (10 comma negative 26)
- The area below and to the right of the boundary is shaded.

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B)



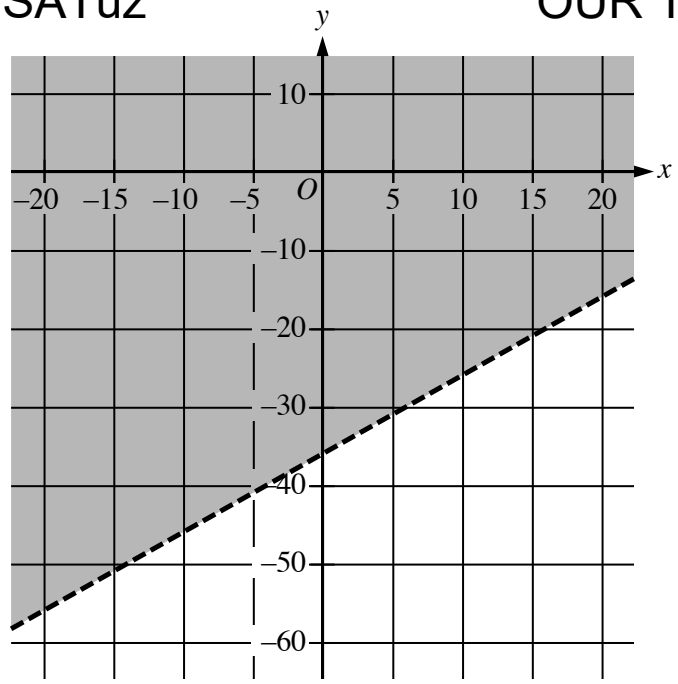
- The boundary of the inequality is a dashed line.
  - The line is horizontal.
  - The line passes through the following points:
    - (negative 10 comma negative 36)
    - (0 comma negative 36)
    - (10 comma negative 36)
- The area above the boundary is shaded.

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C)



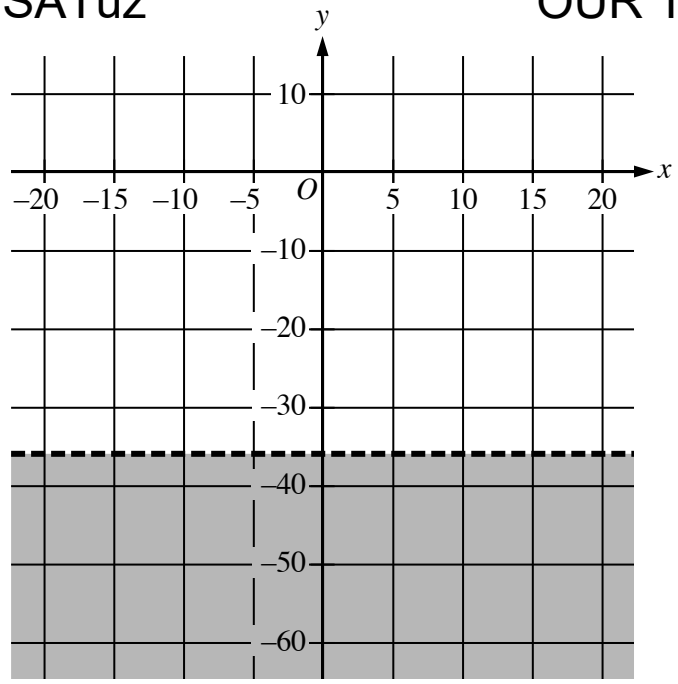
- The boundary of the inequality is a dashed line.
  - The line slants gradually up from left to right.
  - The line passes through the following points:
    - (negative 10 comma negative 46)
    - (0 comma negative 36)
    - (10 comma negative 26)
- The area above and to the left of the boundary is shaded.

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D)



- The boundary of the inequality is a dashed line.
  - The line is horizontal.
  - The line passes through the following points:
    - (negative 10 comma negative 36)
    - (0 comma negative 36)
    - (10 comma negative 36)
- The area below the boundary is shaded.

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Question 2

A chemist combines 25 grams of compound A, 35 grams of compound B, and x grams of compound C. The resulting mixture has a mass of 125 grams. What is the value of x?

A) 25

B) 65

C) 35

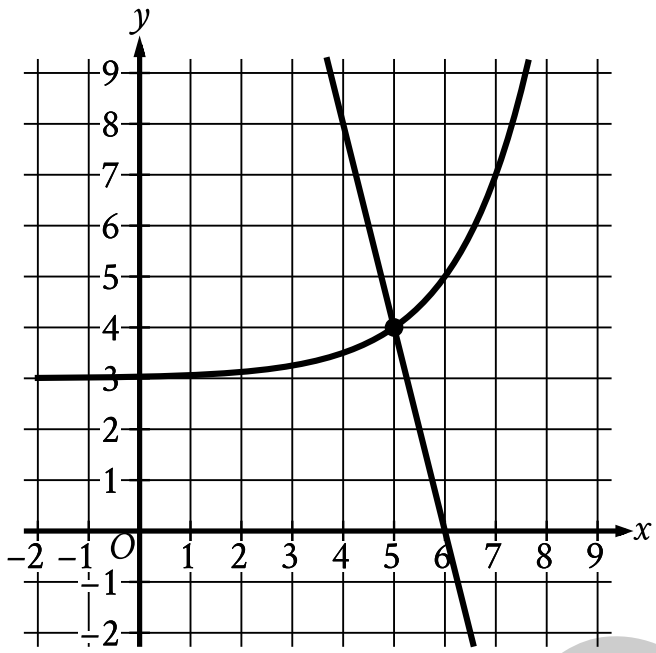
D) 10

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Question 3



- For the curve in the system:
  - The curve passes from quadrant 2 to quadrant 1.
  - In quadrant 2, the curve trends up gradually to the approximate point (0 comma 3.031).
  - In quadrant 1:
    - The curve trends up gradually to point (5 comma 4).
    - The curve then trends up sharply.
  - The curve passes through the following points:
    - (4 comma seven halves)
    - (5 comma 4)

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- (6 comma 5)
- For the line in the system:
  - The line slants sharply down from left to right.
  - The line passes through the following points:
    - (4 comma 8)
    - (5 comma 4)

The graph of a system of a linear equation and a nonlinear equation is shown. What is the solution (x, y) to this system?

A) $(4, 0)$

B) $(5, 4)$

C) $(0, 0)$

D) $(0, 5)$

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Question 4

Carmen needs at least 100 signatures to submit a petition. If Carmen already has 51 signatures, what is the minimum number of additional signatures Carmen needs to submit the petition?

A) 41

B) 151

C) 49

D) 51

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Question 5

There are a total of 340 seats in a school auditorium. During an assembly, students occupied 50% of the seats in the auditorium. How many seats did the students occupy during this assembly?

A) 290

B) 25

C) 170

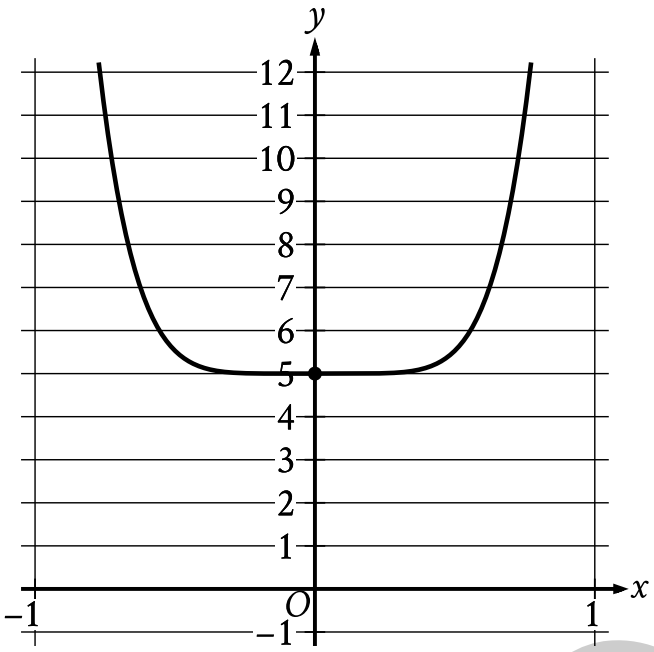
D) 50

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Question 6



- In quadrant 2, the curve falls to its minimum at the point (0 comma 5).
- In quadrant 1, the curve then rises.

The graph of the polynomial function f, where $y = f(x)$, is shown. The y-intercept of the graph is $(0, y)$. What is the value of y?

Enter response...

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Question 7

If $5c = 15$, what is the value of $5c - 8$?

A) 3

B) 67

C) 7

D) 23

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Question 8

The ratio of k to p is constant. If $k = 7$, then $p = 6$. What is the value of k when $p = 336$?

A) 288

B) 56

C) 48

D) 392

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Question 9

Each of 185 animals can be classified as one of three species, as shown in the frequency table.

Classification	Frequency
species X	112
species Y	3
species Z	70

If one of the animals is selected at random, what is the probability of selecting an animal of species Y?

A) $\frac{70}{185}$

B) $\frac{3}{185}$

C) $\frac{182}{185}$

D) $\frac{112}{185}$

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Question 10

For certain air temperatures, the table gives wind chill temperatures for two different wind speeds.

Air temperature	Wind chill temperature at wind speed 5 mph	Wind chill temperature at wind speed 15 mph
27°F	21°F	15°F
29°F	24°F	18°F
31°F	26°F	20°F

According to the table, what is the wind chill temperature, in degrees Fahrenheit $(^\circ\text{F})$, when the air temperature is 29°F and the wind speed is 15 miles per hour (mph)? (Disregard the degree symbol when entering your answer.)

Enter response...

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Question 11

In a right triangle, the measure of one of the acute angles is 14°. What is the measure of the other acute angle, in degrees? (Disregard the degree symbol when entering your answer.)

Enter response...

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Question 12

Which expression is a factor of $3x^2 + 57x + 15$?

A) $3x^2$

B) $5x$

C) 3

D) $57x$

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Question 13

A city has an area of x square miles and a population density of $1{,}035$ people per square mile. The suburb adjacent to the city has an area of y square miles and a population density of 535 people per square mile. The combined population of the city and the suburb adjacent to the city is $13{,}165$ people. Which equation represents this situation?

A) $1{,}035x + 535y = 13{,}165$

B) $1{,}035x + 535 = 13{,}165y$

C) $1{,}035x + 535 = 1{,}570y$

D) $1{,}035x + 535y = 1{,}570$

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Question 14

A triangle has a base length of 48 centimeters and a height of 96 centimeters. What is the area, in square centimeters, of the triangle?

A) $2{,}304$

B) $1{,}152$

C) $4{,}608$

D) 144

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Question 15

The function $f(x) = -40x + 340$ gives the predicted height above the ground $f(x)$, in feet, of a model airplane x minutes after it begins to descend. What is the predicted height above the ground, in feet, of the model airplane 3 minutes after it begins to descend?

A) 120

B) 220

C) 300

D) 460

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Question 16

$$5a = \frac{4b^2}{a} - 20a$$

The given equation relates the positive variables a and b. Which equation correctly expresses a in terms of b?

A) $a = \frac{4b}{25}$

B) $a = \frac{4b^2}{25}$

C) $a = \frac{2b}{5}$

D) $a = \frac{2b^2}{5}$

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Question 17

Charles saves $\frac{3}{5}$ of the $\$225$ he earns each week from his summer job. If Charles continues to save at this rate, how much money, in dollars, will Charles save in 3 weeks?

Enter response...

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Question 18

The function f is defined by $f(x) = 2x^3 + \frac{1}{3}x^2 - x$. What is the value of $f(1)$?

A) $\frac{4}{3}$

B) 2

C) $\frac{10}{3}$

D) 6

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Question 19

In the *xy*-plane, what is the slope of the line that passes through the points $(0, 0)$ and $(13, 14)$?

A) 13

B) $\frac{14}{13}$

C) $\frac{13}{14}$

D) 14

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Question 20

Function f is defined as $f(x) = (2x - 5)^2$, and function g is defined as $g(x) = 2(2x + 7)$. Which of the following is equivalent to $f(x) + g(x)$?

A) $4x^2 - 16x + 11$

B) $4x^2 - 16x + 39$

C) $4x^2 - 6x + 39$

D) $4x^2 + 24x + 11$

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Question 21

$$15x^2 = 15(9)$$

What is the negative solution to the given equation?

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Question 22

The function f is defined by $f(x) = a(4)^x$, where a is a positive constant. In the *xy*-plane, the graph of $y = f(x)$ passes through the point $(0, 13)$. What is the value of $f(0)$?

A) 4

B) 1

C) 13

D) 9

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